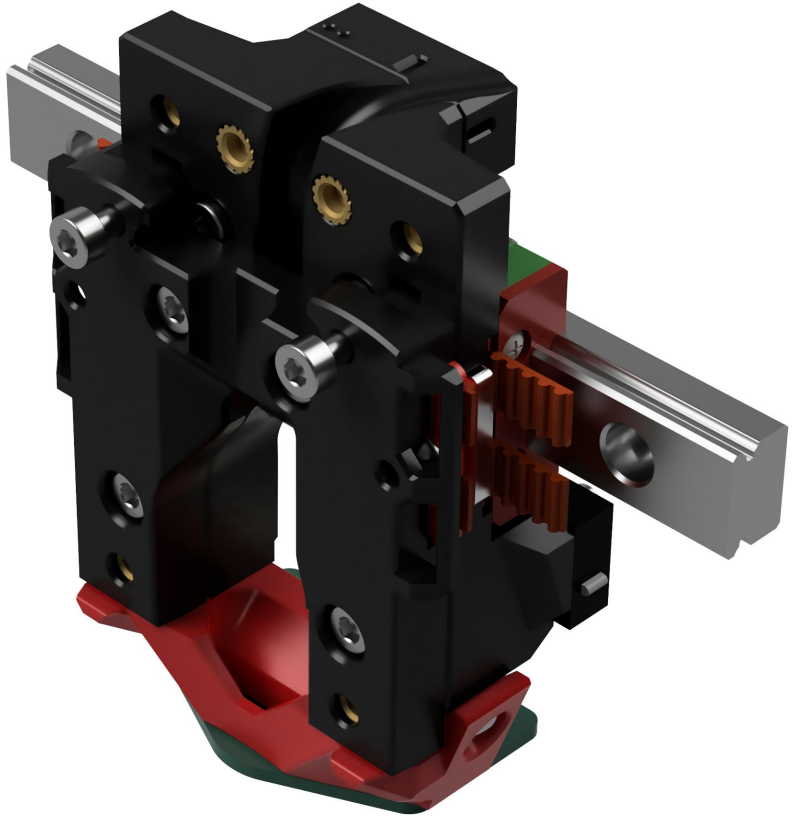
VORON



VORON CHARIOT ASSEMBLY

We build space shuttles with gardening tools
so anyone can have a space shuttle of their own.

VERSION 2026-10-26



Before you begin on your journey, a word of caution:

You are about to assemble a robot in the comfort of your own home. This machine can maim, burn, and electrocute you if you are not careful. Please do not become the first VORON fatality. There is no special Reddit flair for that.

Please read the entire manual before you start assembly. As you begin wrenching, please check our Discord channels for any tips and questions that may halt your progress.

Most of all, good luck!

THE VORON TEAM

Introduction 04

Hardware Reference 07

Main Carriage 10

Probe Mount 16

Belt Clips 21

Front Plate 28

Finishing Touches 30

PART PRINTING GUIDELINES

The VORON Team has provided the following print guidelines to follow for the best chance at success with your parts. There are often questions about substituting materials or changing printing standards, but we recommend that you follow these:

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS or ASA

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 40%

WALL COUNT

Recommended: 4

SOLID TOP/BOTTOM LAYERS

Recommended: 5

PRINT IT FORWARD (PIF)

Often times community members that have issues printing ABS will bootstrap themselves into a VORON using our Print It Forward program. This is a service where approved members with VORON printers can make you a functional set of parts to get your own machine up and running. Check Discord if you have any interest in having someone help you out.

FILE NAMING

By this time you should have already downloaded our STL files from the VORON GitHub. You might have noticed that we have used a unique naming convention for the files. This is how to use them.

PRIMARY COLOR

Example z_joint_lower_x4.stl

These files will have nothing at the start of the filename.

ACCENT COLOR

Example [a]_tensioner_left.stl

We have added “[a]” to the front of any STL file that is intended to be printed with accent color.

QUANTITY REQUIRED

Example: [a]_z_belt_clip_lower_x4.stl

If any file ends with “_x#”, that is telling you the quantity of that part required to build the machine.

HOW TO GET HELP

If you need assistance with your build, we’re here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck.

DISCORD https://discord.gg/voron

VORON FORUM https://forum.vorondesign.com

REPORTING ISSUES

Should you find an issue in the documentation or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Chariot/issues).
When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome. We periodically update the manual based on the feedback we get.

THIS IS JUST A REFERENCE

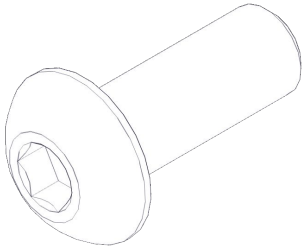
This manual is designed to be a simple reference manual. Building a VORON can be a complex endeavour and for that reason we recommend downloading the CAD files off our Github repository if there are sections you need clarification on. It can sometimes be easier to follow along when you have the whole assembly in front of you.



https://github.com/vorondesign



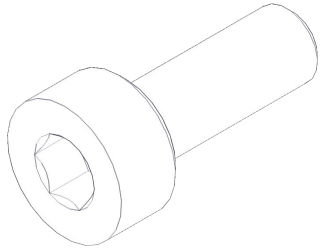
https://docs.vorondesign.com



BUTTON HEAD CAP SCREW (BHCS)

Metric fastener with a domed shape head and hex drive. Most commonly found in locations where M5 fasteners are used.

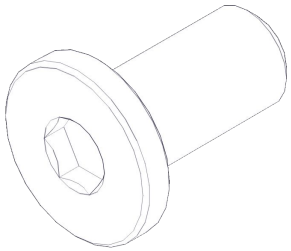
ISO 7380-1



SOCKET HEAD CAP SCREW (SHCS)

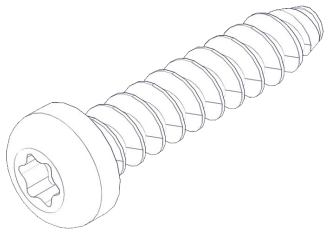
Metric fastener with a cylindrical head and hex drive. The most common fastener used on the VORON.

ISO 4762



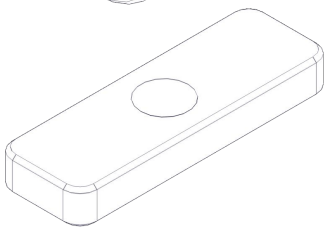
ULTRA LOW PROFILE SHCS

Metric fastener with a reduced-height cylindrical head and hex drive. Used where standard SHCS heads would interfere with nearby components.



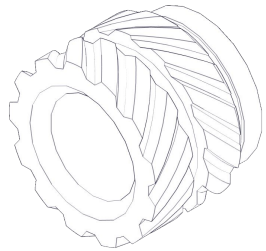
SELF TAPPING SCREW

Fastener with a pronounced thread profile that is driven directly into plastic.



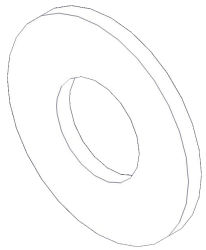
MAKERBEAM T-SLOT NUT

Nut designed to slide into the T-slot channel of a MakerBeamXL extrusion. Tightening the screw pulls the nut against the inside of the slot, clamping it in place.



HEAT SET INSERT

Use a soldering iron to heat the insert and melt it into the plastic as it's pressed in. As the plastic cools, it holds on to the knurls and ridges, giving excellent resistance to both torque and pull-out.



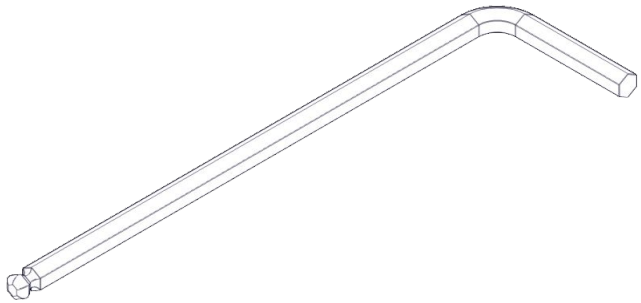
WASHER

Usually stamped from sheet metal this type of spacer is not as consistent in thickness as the shims are. Only used in M3 size.

DIN 125

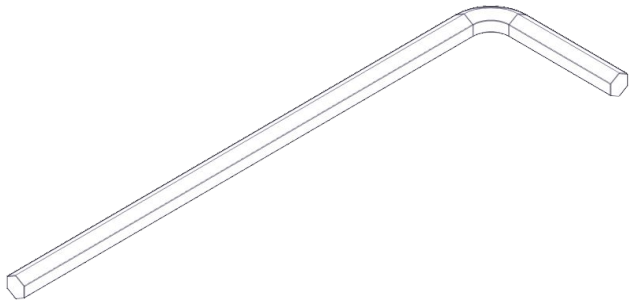
BALL-END HEX DRIVER

Some parts of this build require a ball-end hex driver. We recommend having 2.0 and 2.5mm sizes available.



HEX DRIVER

The 2.0 and 2.5mm hex drivers see heavy use in this build. Invest in quality drivers. Refer to the sourcing guide for suggestions.

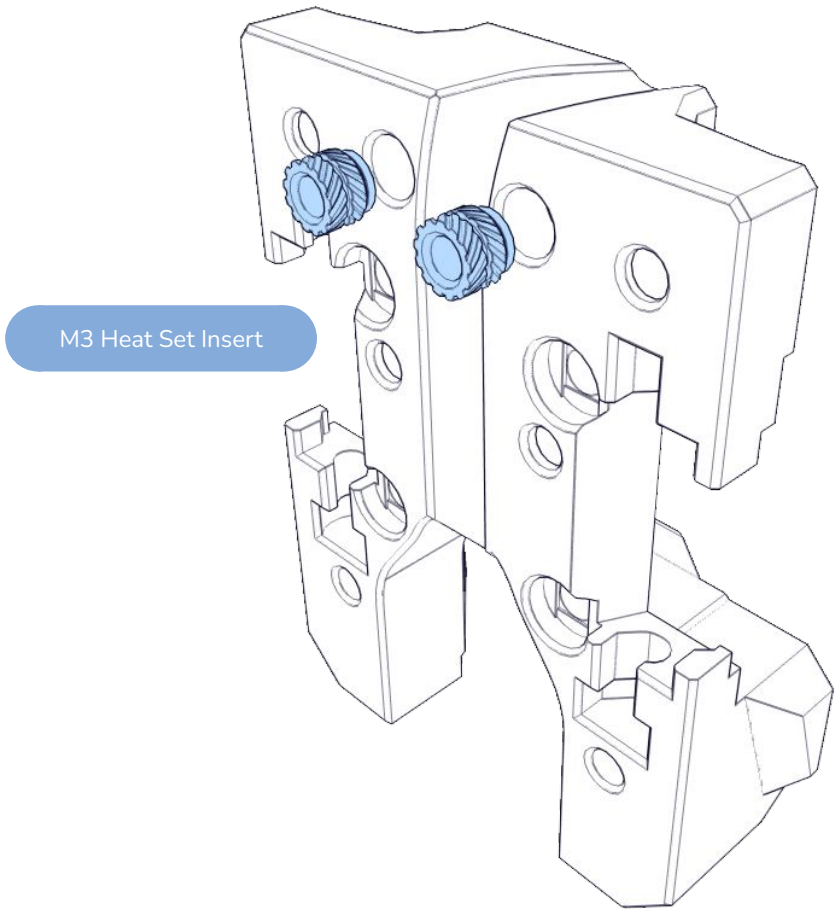


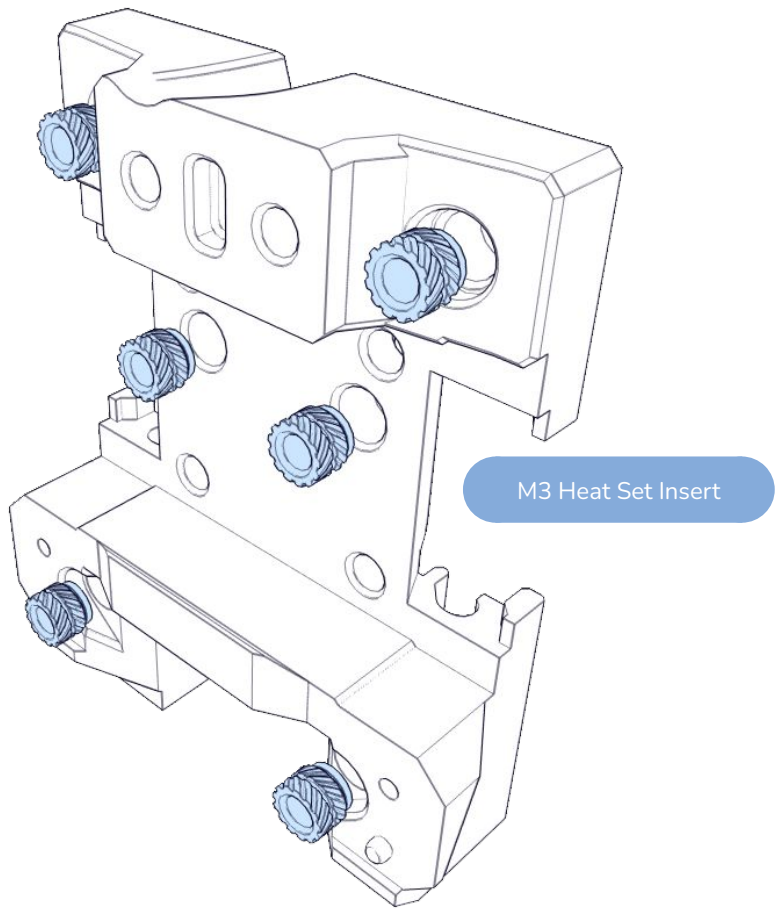
ADDITIONAL TOOLS

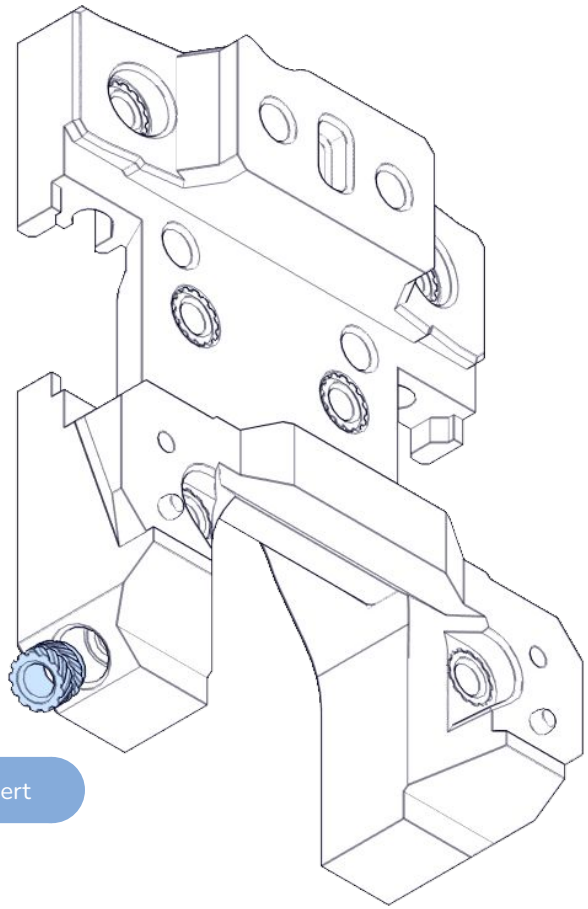
Additional tool recommendations are available in the sourcing guide. Visit https://vorondesign.com/sourcing_guide and select the "VORON Tools" tab at the bottom of the page.

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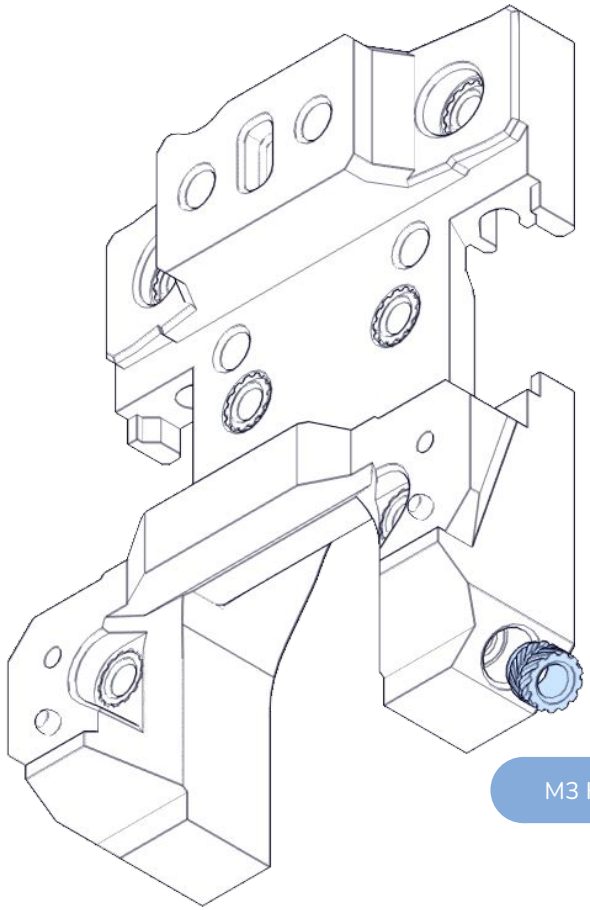
9


M3 Heat Set Insert

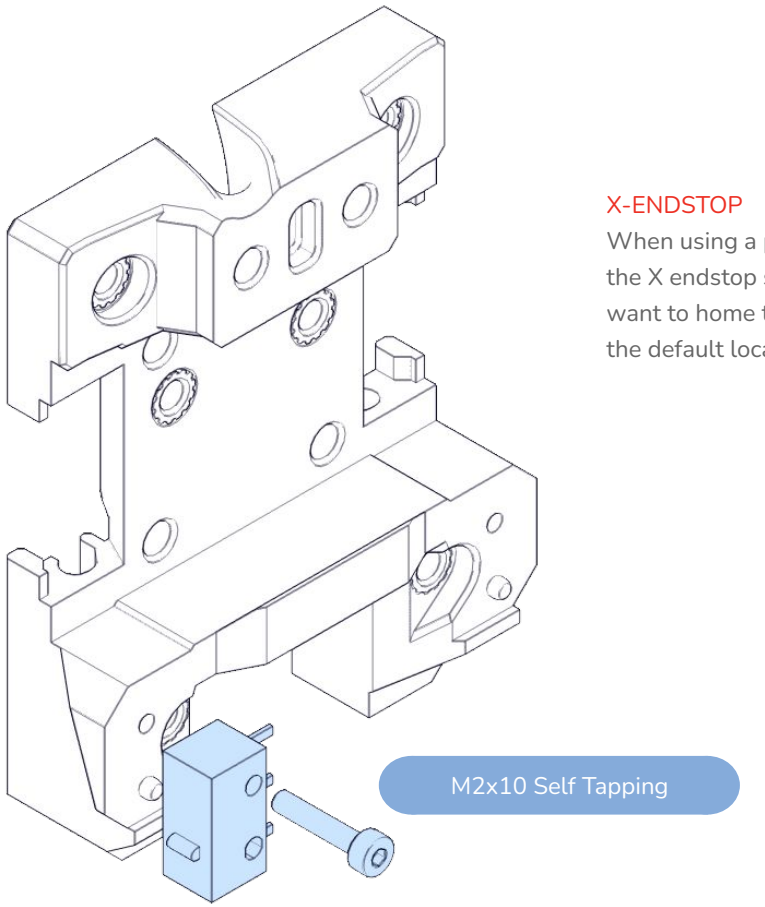

M3 Heat Set Insert



M3 Heat Set Insert

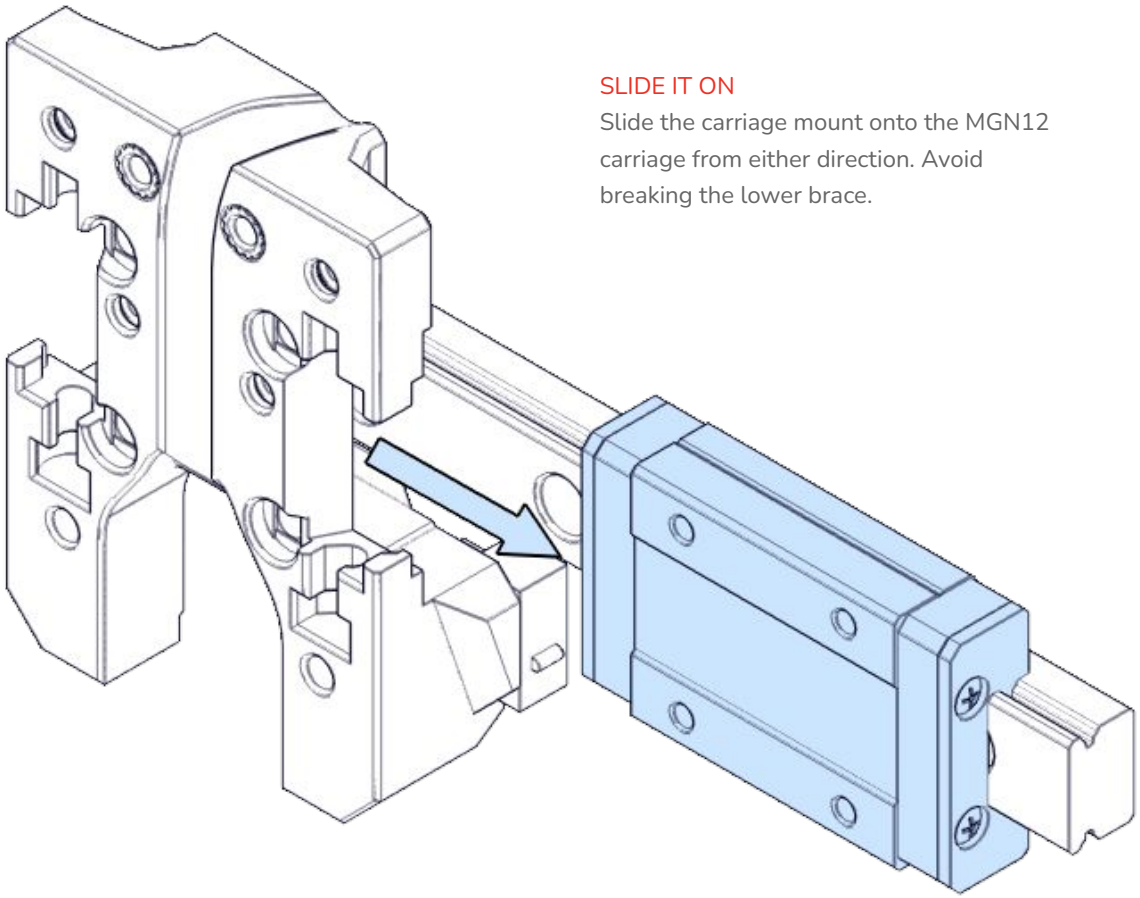


M3 Heat Set Insert



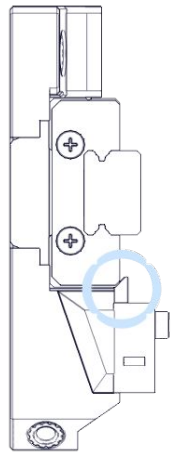
X-ENDSTOP

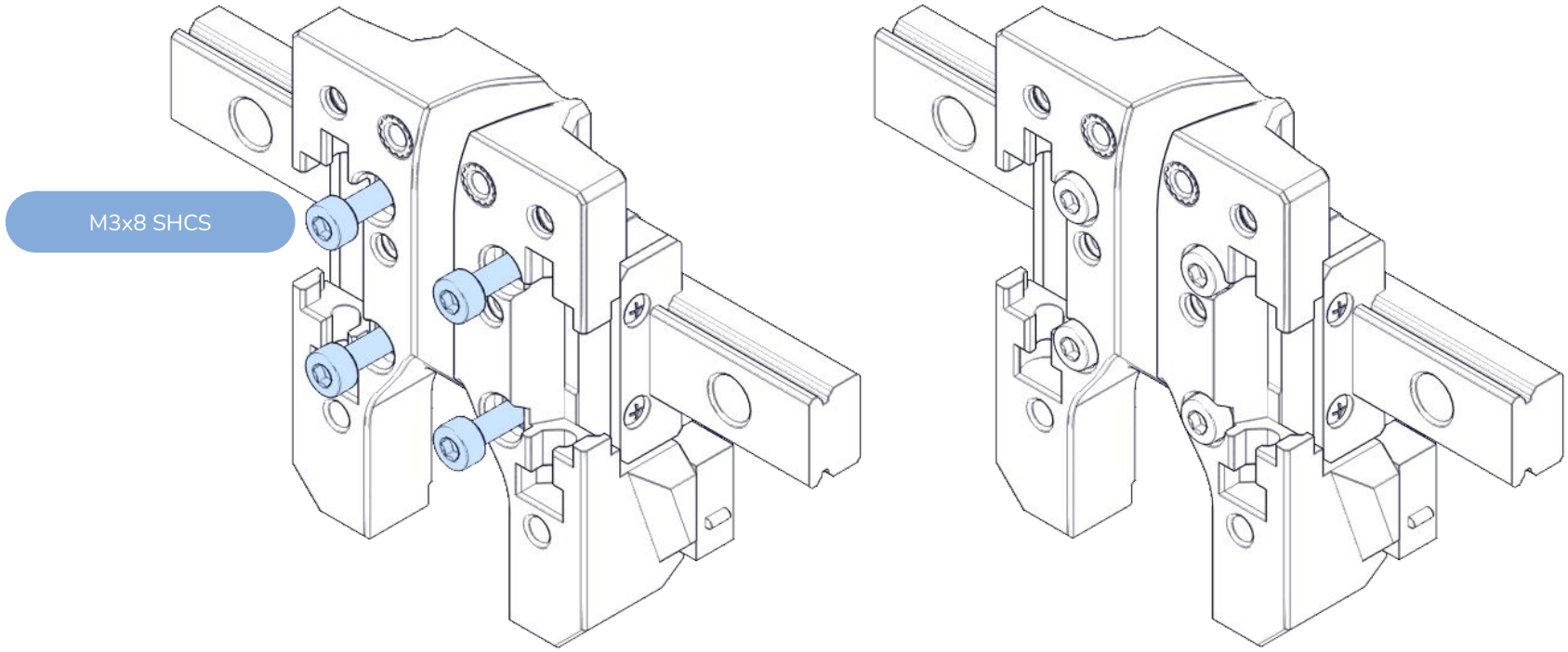
When using a physical endstop, install the X endstop switch on the side you want to home to. The position shown is the default location for homing to X-max.

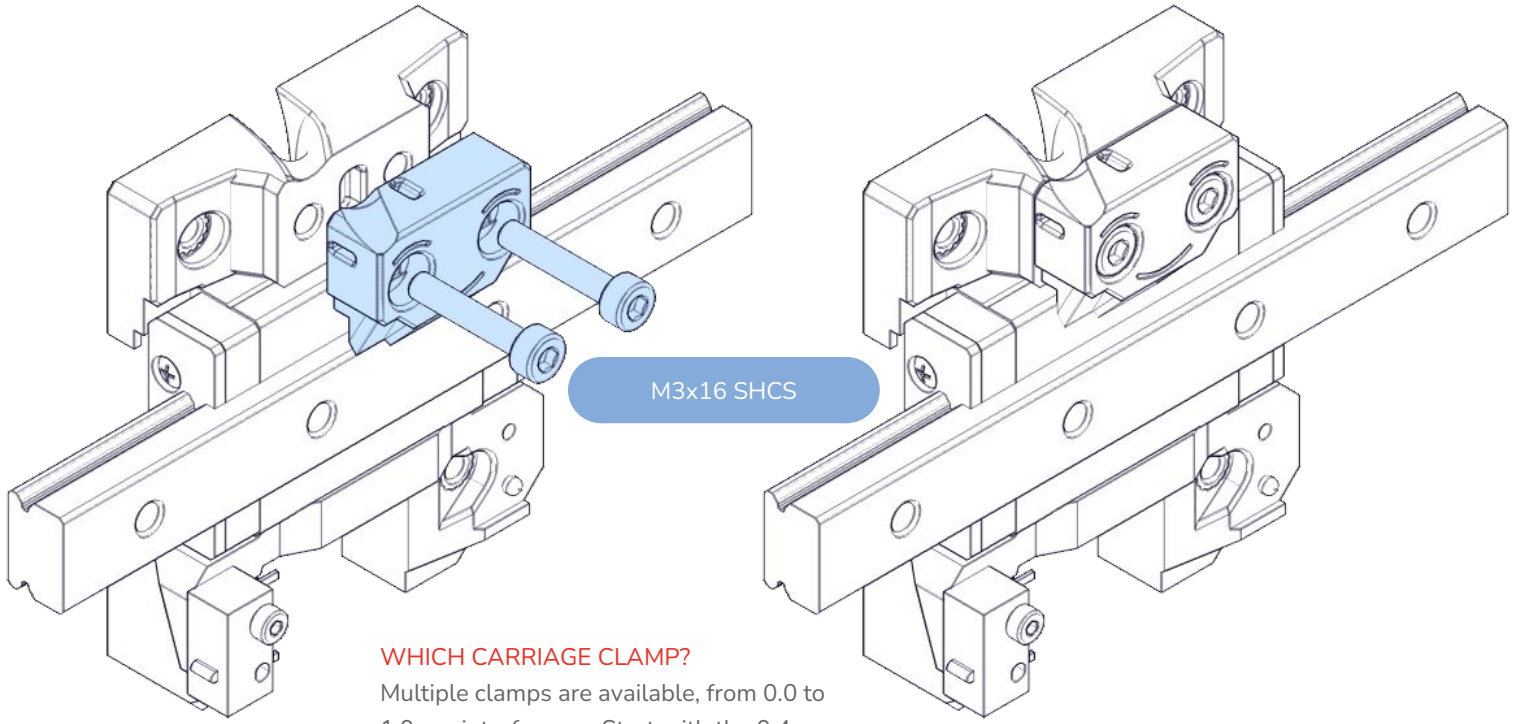


SLIDE IT ON

Slide the carriage mount onto the MGN12 carriage from either direction. Avoid breaking the lower brace.




M3x8 SHCS



WHICH CARRIAGE CLAMP?

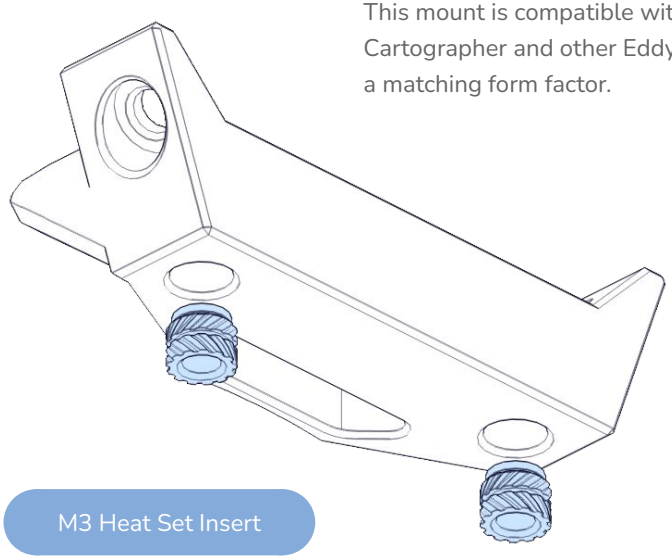
Multiple clamps are available, from 0.0 to 1.0mm interference. Start with the 0.4mm clamp, then try others to see if they improve your Input Shaper results.Too much interference can cause spikes in your graphs.

PICK ONE

You can choose from multiple probe mount parts, depending on what kind of probe you intend to use. You only need **one** of the options shown here. The manual will show a eddy current probe being used.

EDDY CURRENT PROBE

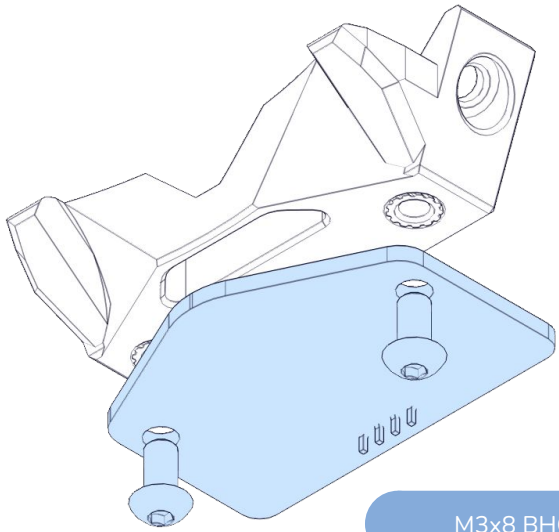
This mount is compatible with Beacon, Cartographer and other Eddy probes with a matching form factor.



M3 Heat Set Insert

DON'T SCREW IT UP

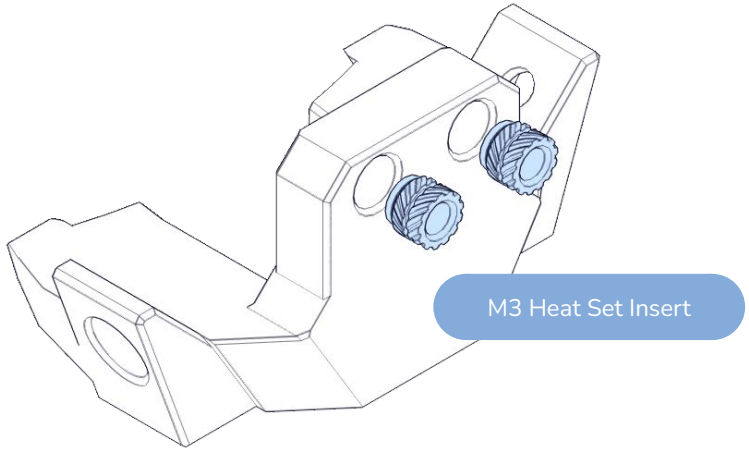
If your probe comes with low-profile screws, use those instead of the M3x8 BHCS called out here.

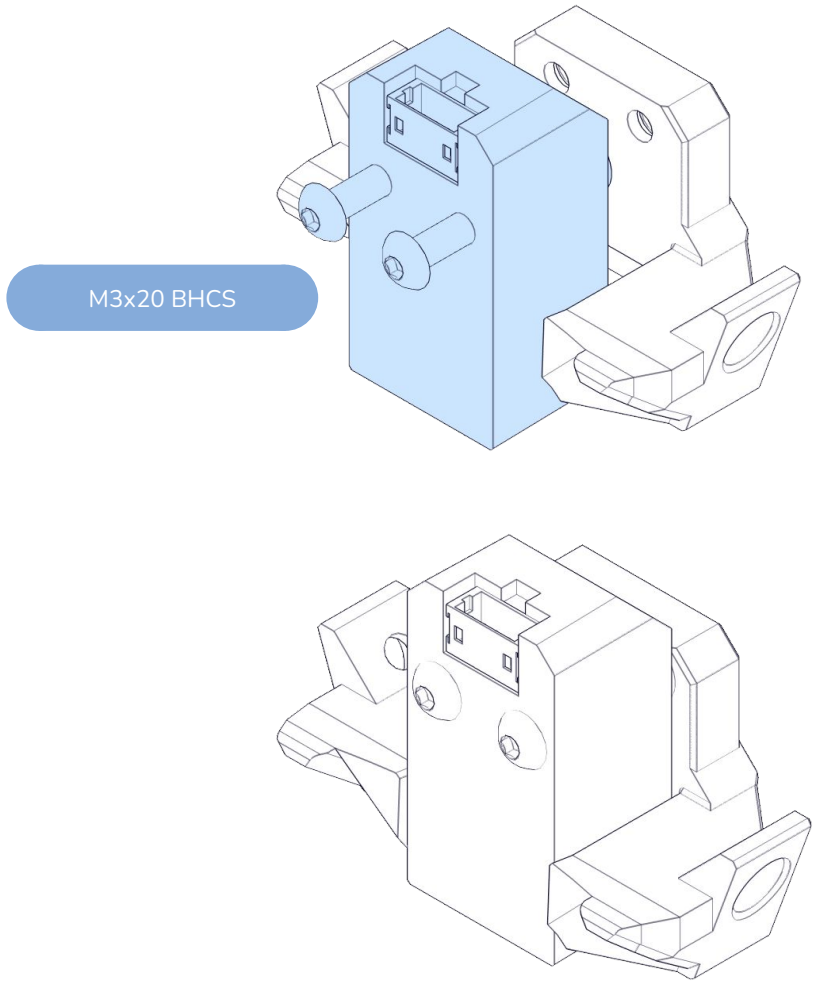


M3x8 BHCS

ONLY ONE - BTT EDDY PROBE

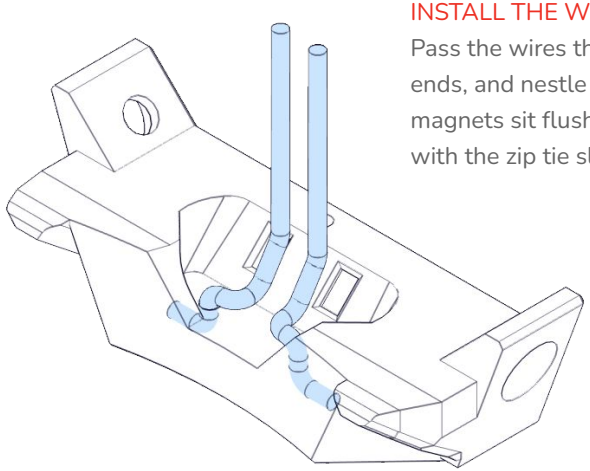
You only need one of the probe mount options. Skip this page if you're not using a BTT Eddy probe.





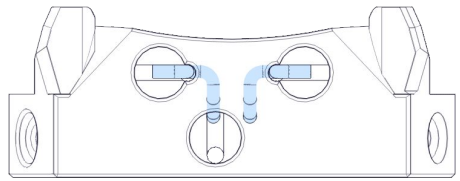
ONLY ONE - KLICKY PROBE

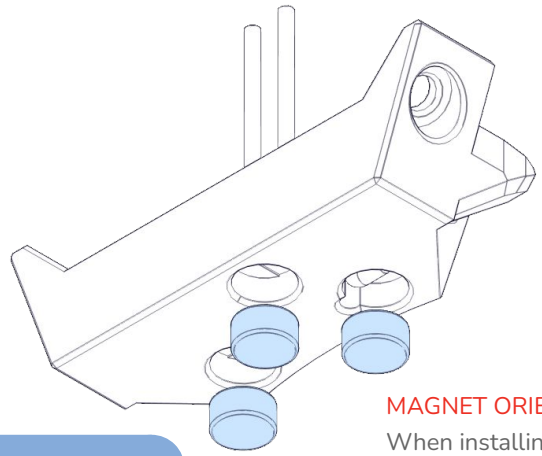
You only need one of the probe mount options. Skip this page if you're not using a Klicky style dockable probe.



INSTALL THE WIRES

Pass the wires through the cable channels, strip the ends, and nestle them into the grooves so the magnets sit flush with the surface. Secure the wiring with the zip tie slot provided.

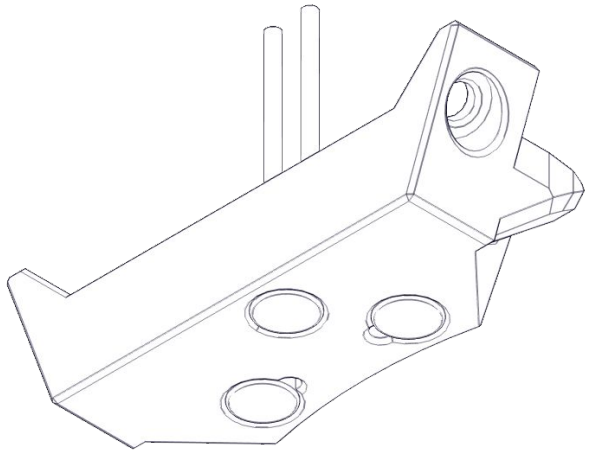




6x3 mm Magnets

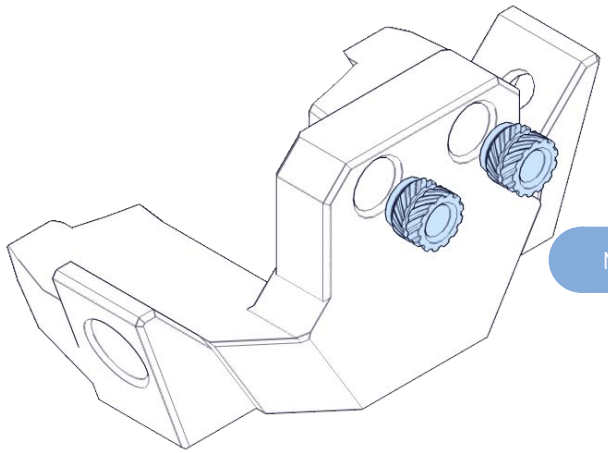
MAGNET ORIENTATION

When installing the magnets, make sure the north/south pattern matches your attachable probe.

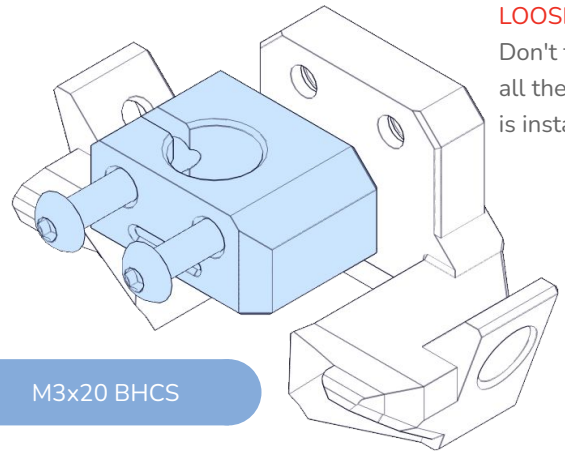


ONLY ONE - PINDA PROBE

You only need one of the probe mount options. Skip this page if you're not using a Pinda style probe.



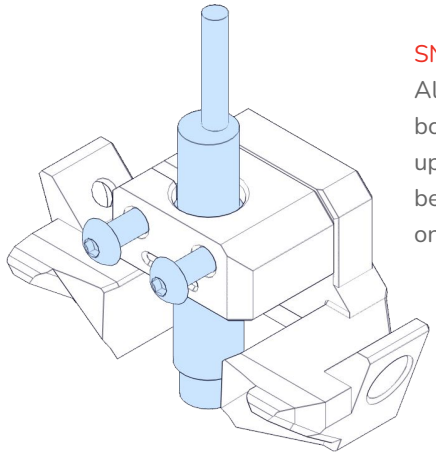
M3 Heat Set Insert



M3x20 BHCS

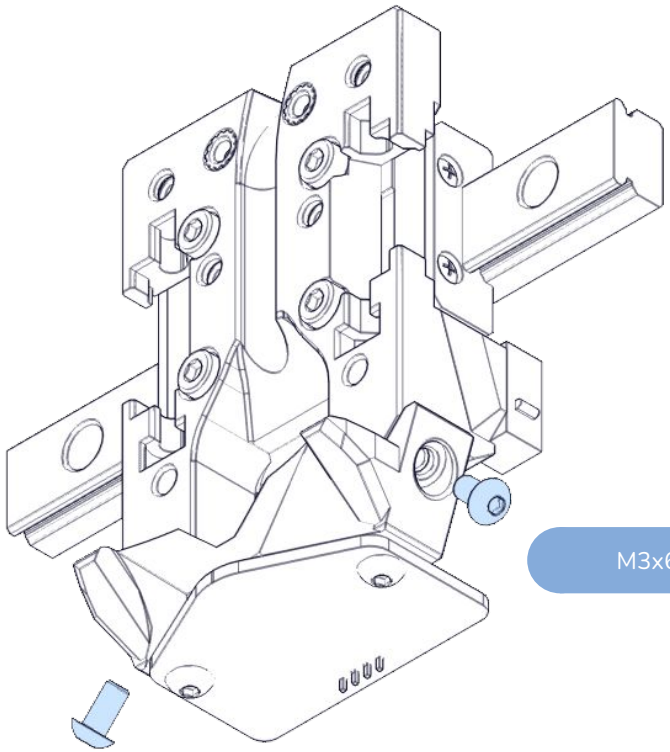
LOOSE FOR NOW

Don't tighten the M3x20 screws all the way. Wait until the probe is installed.



SNUG IT UP

Align the end of the probe with the bottom surface of the mount and snug up the screws. Some fine-tuning of the bed-to-probe distance may be required once the toolhead is installed.

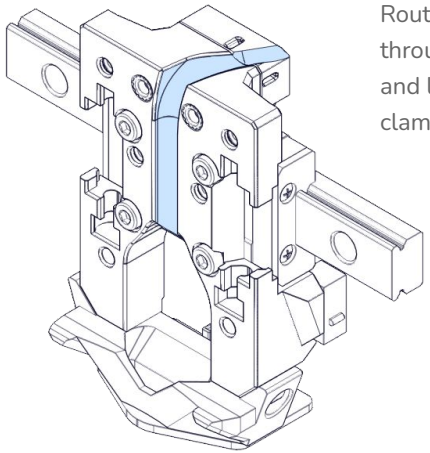


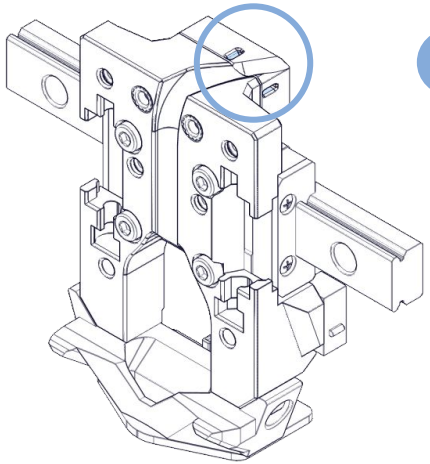
M3x6 BHCS

M3x6 BHCS

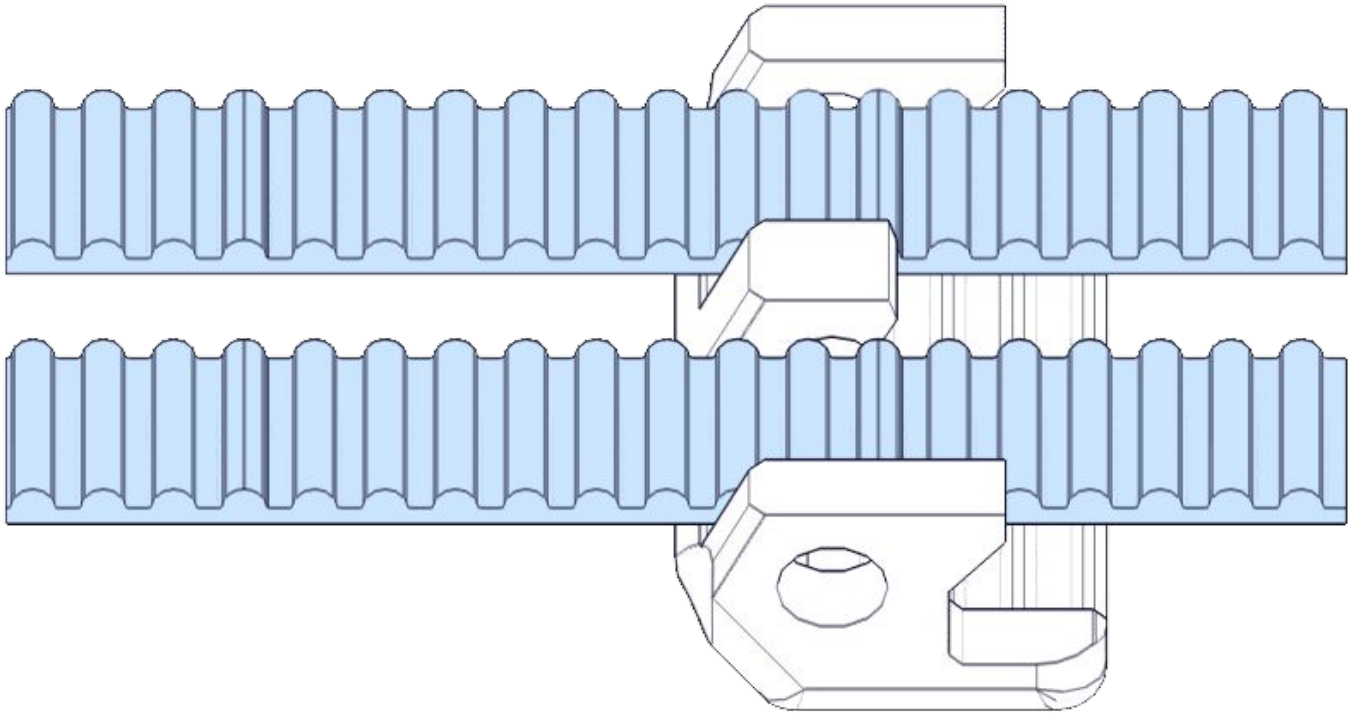
WIRE MANAGEMENT

Route the probe and endstop wires through the highlighted cable channel and loosely secure them to the carriage clamp with a zip tie.





ZIP TIE LOOP



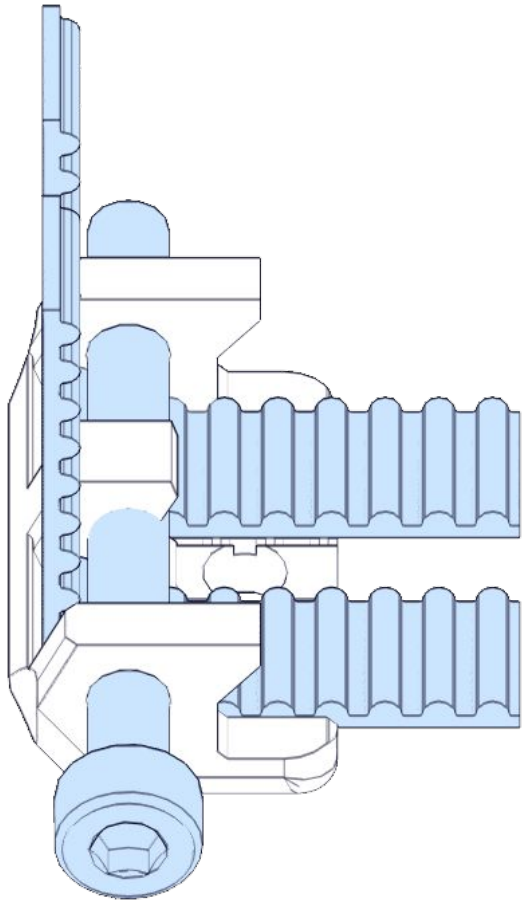
Belt ends

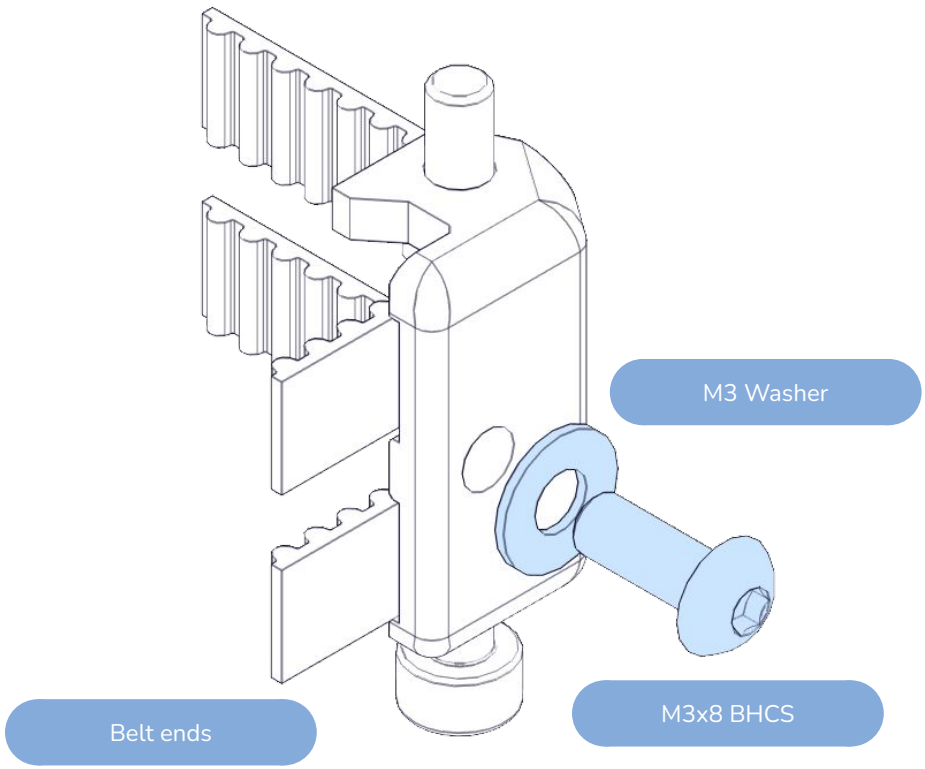
BELT ORIENTATION

Lay the belts in the clip as shown, with the ends facing to the right. This image shows the left-side clip.

SCREW ALIGNMENT

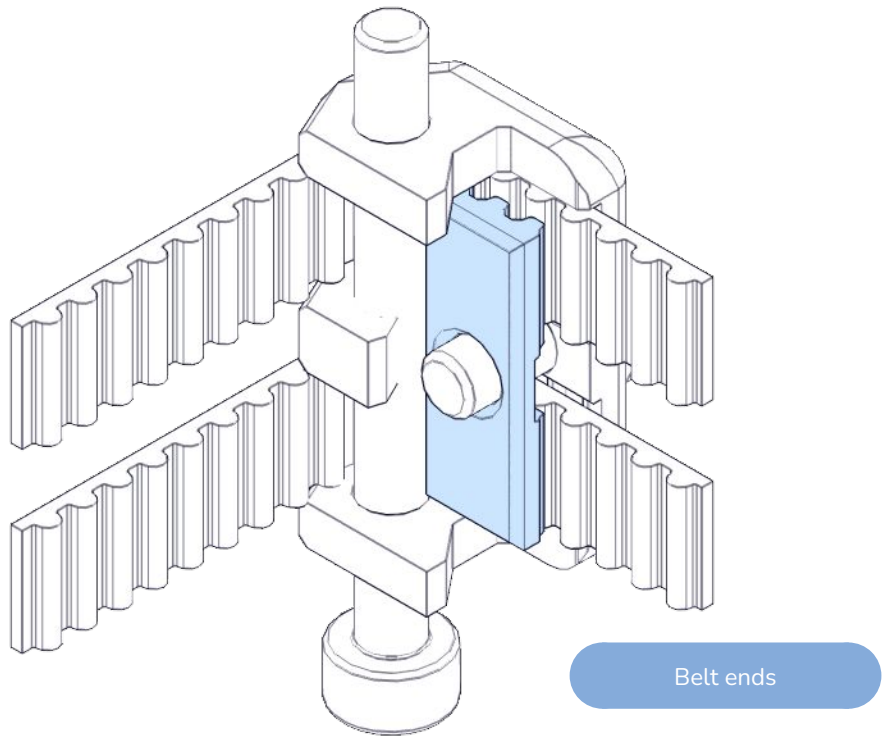
Leave approximately 5 mm between the screw head and the clip. This screw installs through opposite sides for each clip.





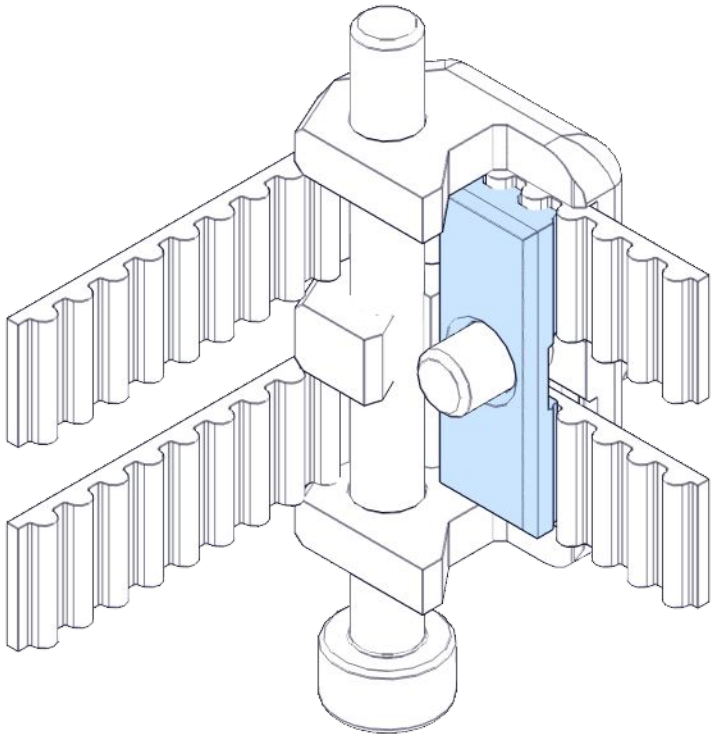
CLAMPING THE BELT DOWN

Loosely install the M3×8 BHS with washer to the other side of the clip.



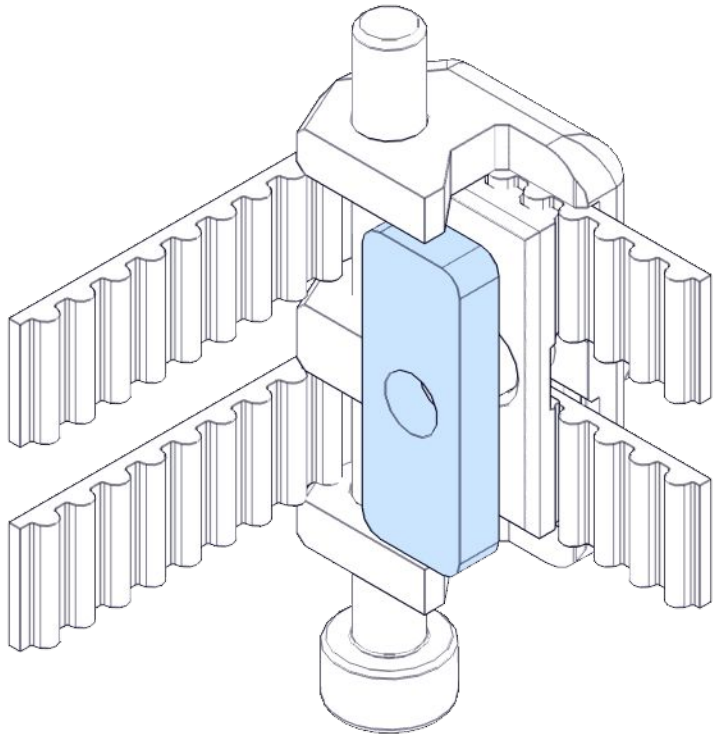
FLAT SIDE OUT

Install the belt clip insert onto the M3x8 screw with the flat side facing away from the belt.



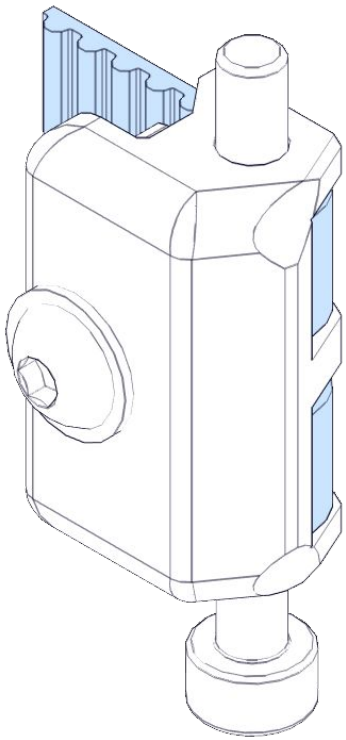
BELT ALIGNMENT

Make sure the ridges on the belt nestle correctly into the grooves of the belt clip insert.



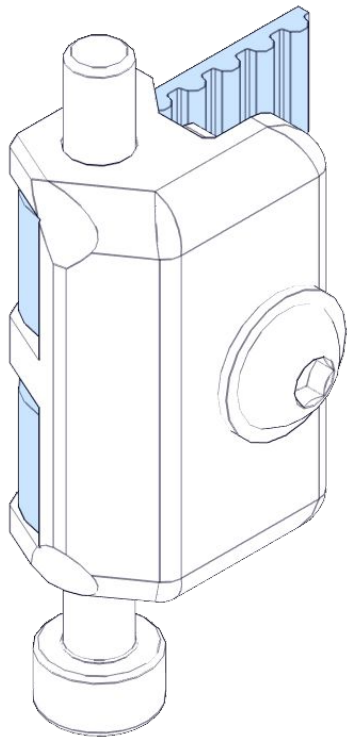
MATCH TOOTH COUNT

Loosely install the MakerBeamXL T-slot nut. Even up the belts so the same tooth count sticks out on both belts, then tighten the M3x8 BHCS.



WRAP THE BELT

Lightly fold the belt around the M3x30 screw until it wraps approximately 180°.

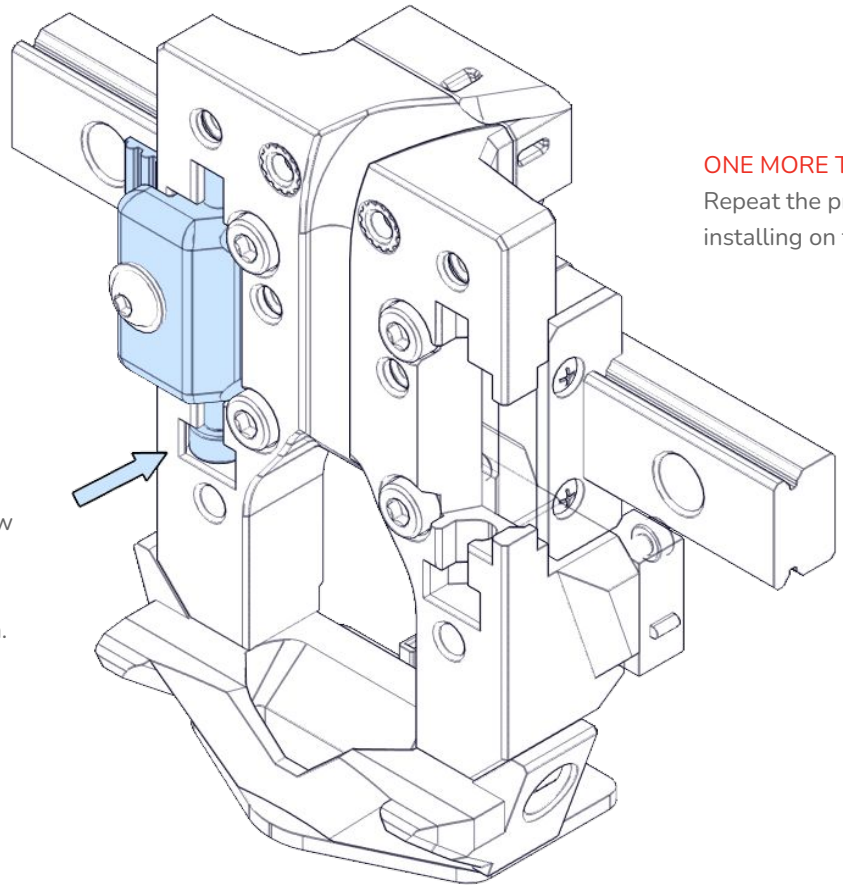


AGAIN, BUT BACKWARDS

Repeat the process for the second belt clip. The process is mirrored: the M3x30 screw installs from the opposite side.

INSTALL THE CLIP

Place the T-slot nut side of the clip into the carriage, then roll the belt clip in until the M3x30 screw is captured by the carriage mount.



ONE MORE TIME

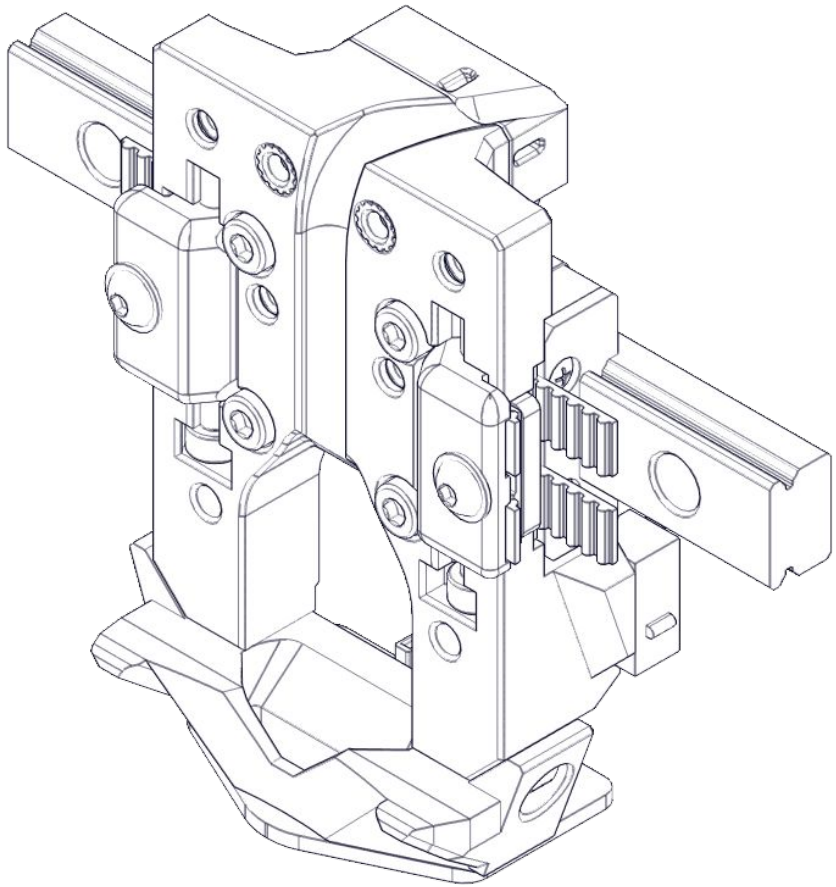
Repeat the process for the other belt clip, installing on the opposite side.

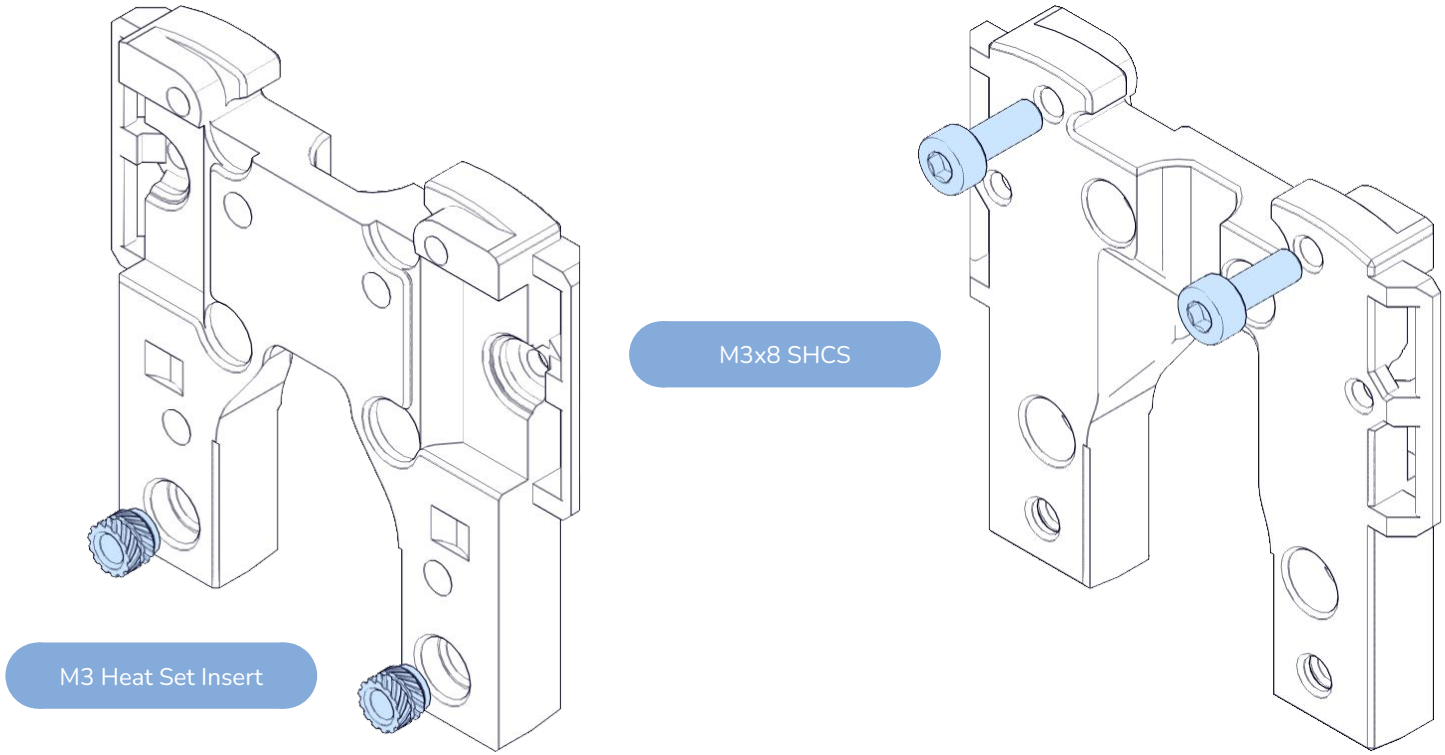
ADJUST THE SCREW

You may need to adjust the M3x30 screw to get the clip to install easily. No force should be required. If there's too much resistance, check where it's coming from.

CHECK YOUR WORK

Before moving on to install the front plate, make sure everything is properly installed.



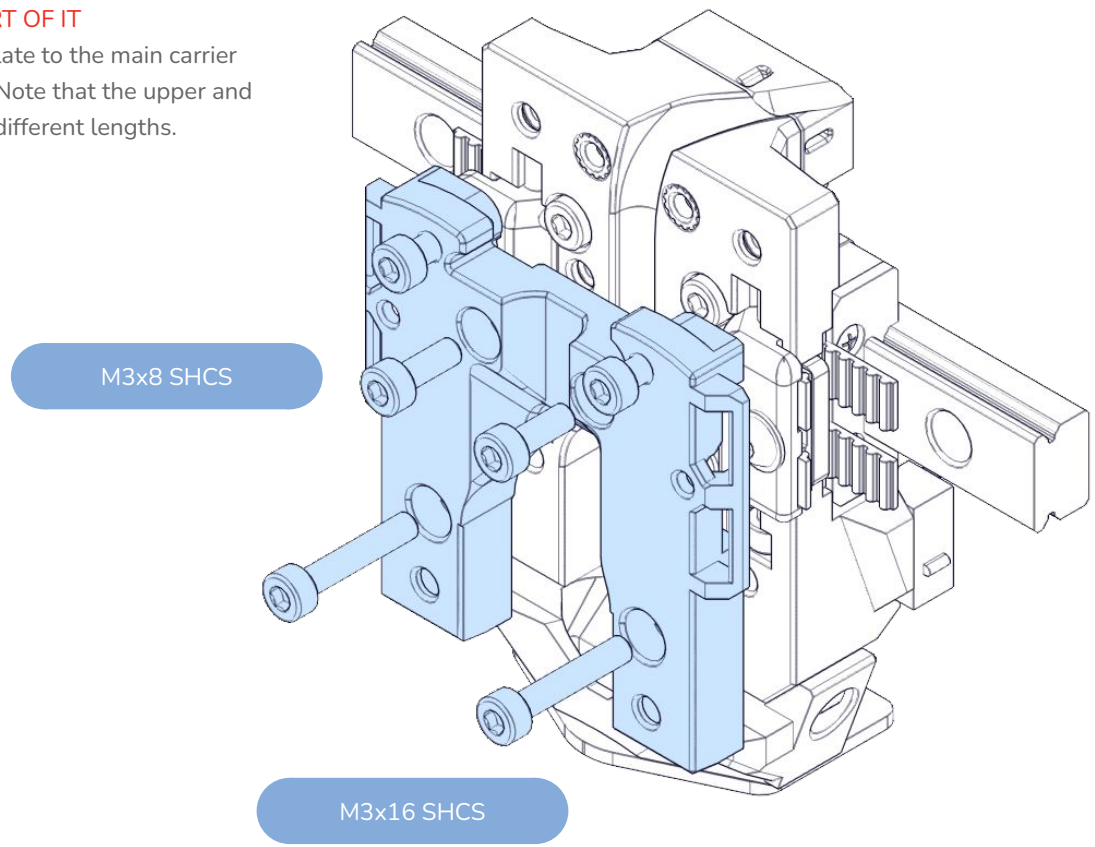


TOOLHEAD DEPENDENT

Stealthburner and A4T require these two screws. Other toolheads may differ. Leave about 3 mm between the screw head and the plastic.

LONG AND SHORT OF IT

Attach the front plate to the main carrier with four screws. Note that the upper and lower screws are different lengths.



WIRE MANAGEMENT

Snug up the zip tie holding the probe and endstop wires. Position the zip tie head so it doesn't interfere with the extruder or toolhead board.

CHECK THE NOZZLE TO PROBE CLEARANCE

The probe heights on the included mounts are set up for stock Voron tool heads. If your tool head is different, you need to check the clearance before homing. If you need to adjust the probe height, the native CAD files have been supplied to make your life a little easier.

ADJUSTING THE BELT LENGTH

If you need to adjust belt length after the carriage is assembled, follow this procedure:

1. Loosen one of the belt clip screws a turn or two. You can reach the screws through the access holes in the front plate.
2. Adjust the belt length one tooth at a time to keep both belts even.
3. Tighten the belt clip screw again.
4. If your belts ends are long enough, loop them into the belt retainers on the front plate.

CONGRATULATIONS, ASTRONAUT

Install your toolhead and start printing!



Website	GitHub	Docs	Discord
vorondesign.com	github.com/vorondesign	docs.vorondesign.com	discord.gg/voron

